Jaisurya Sundarbabu

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Education

University of California - San Diego Sep 2025 – Dec 2026 (expected)
Master of Science in Data Science *GPA: 3.77/4.0*
- **Coursework:** Probability, Statistical Modeling, Business Forecasting, Recommender systems, Efficient AI

Anna University, Chennai Jul 2019 – May 2023
B.Tech in Information Technology *GPA: 3.62/4.0*
- **Coursework:** Statistics, Data Structures & Algorithms, Database Systems, Big Data Analytics, Machine Learning

Skills

Languages: Python, C/C++, SQL, JavaScript (JS)
LLM Inference & Serving: LLM inference, KV-cache optimization, quantization, pruning, model compression, on-device/edge inference, latency & throughput benchmarking (p50/p95), Hugging Face Transformers/Diffusers
Machine Learning & Deep Learning: Deep learning, machine learning, NLP, computer vision, supervised fine-tuning, retrieval-augmented generation (RAG), experimentation & hypothesis testing, regression, statistics
Systems & MLOps: PyTorch, TensorFlow, Keras, Scikit-learn, Databricks, Spark/PySpark, Delta Lake, MLflow, AWS, Azure, Git, backend APIs, scalable system design, end-to-end pipelines

Experience

Qualcomm San Diego, CA
Machine Learning Engineer Intern *Jun 2026 – Sep 2026*
- Designed a **sparse attention** method to accelerate the **prefill stage** by selecting the most important KVs, cutting the KV budget from **8,192** to **1,024** tokens with under **5%** PPL degradation.
- Built a **KV-cache eviction** policy enabling a **2,048**-context model to outperform an **8,192**-context HF baseline, improving PPL by **5%** despite the expected quality tradeoff from eviction.
- Researched **activation-aware quantization**, excluding outlier-token activations from the quantization range to improve **SQNR by about 6dB** (about **4x** lower quantization noise) while keeping PPL degradation under **5%**.

FastVideo / Hao AI Lab, UC San Diego San Diego, CA
Graduate Student Researcher *Jan 2026 – Present*
- **Open-source contributor to FastVideo (3K+ GitHub stars)**, integrating **Kandinsky-5.0 Video Lite** by adding DiT configs and wiring model registration/loading for Diffusers-style checkpoints.
- Built **numerical parity** and **regression tests** to validate weight mapping and match transformer outputs vs. the reference implementation, unblocking a draft PR toward **T2V Lite pipeline** support.

Celebal Technologies Jaipur, India
Associate Data Scientist *Mar 2024 – May 2025*
- Built an **end-to-end LLM-based RAG** system with **semantic search**(retrieval) for manufacturing issue resolution, and achieved **95%** accuracy on unseen issues, reducing new tickets raised by **70%**.
- Developed **predictive models** for retail **demand forecasting** on large historical datasets using time-series feature engineering and **statistics**-driven backtesting; delivered **15% MAPE** and supported planning decisions.
- Automated retail cataloging via a **Databricks** pipeline over **Delta tables** (photoshoot images), using a **computer vision** classifier and **prompt-based** attribute extraction to generate e-commerce-ready data.

Samsung Digital Commerce Chennai, India
Software Development Engineer *Jul 2023 – Jan 2024*
- Owned **backend** services for the Samsung Shop app, designing and integrating **6 production APIs** and automating recurring jobs to keep data fresh, improving system availability and **reliability** at scale.
- Designed, developed, and tested backend changes for the **global Samsung S24 launch**, implementing **logging and validation checks** and access controls aligned to internal compliance and production standards.

Projects

Adaptive KV-Cache Admission for LLM Serving
- Built **DIP-KV**, an adaptive **KV-cache admission** policy for **vLLM**'s CPU offload tier (probation insertion, ghost list, sampled shadow caches) via vLLM's plugin API. On **Kimi production traces** on an **A100**, cut **TTFT** vs. vLLM's LRU by **40%/25%** on agentic traffic at 43.6% hit rate (**p50/p95**: 1209/4550ms) and 15%/13% on chat.
- Showed **LRU**, **LFU** and cost-aware eviction perform identically at GPU-memory capacity since 85% of KV chunks are never reused, found vLLM's **ARC** policy degrades to LRU, and built a trace-driven simulator that predicts measured GPU hit rates within 0.7 percentage points.

Edge AI Optimization Study: Quantization & Pruning
- Applied post-training **quantization** (to 8-bit) and structured **pruning** (70% sparsity) to a deep learning model in **PyTorch**, retaining 90% accuracy with a 40% reduction in **latency/throughput** gain of 1.6x.